
■ ■ NIDS — AI Network Intrusion Detection Forensic Analysis Report

Session: 2026-09-20 19:59:56 — 2026-09-20 20:01:05 | Host: Binoy | Mode: Simulate

Executive Summary

■■ THREATS DETECTED — Immediate investigation required.

Metric	Value
Total Flows Analysed	103
Clean Flows	0
■ Critical Threats	103
■ High Threats	0
■ Medium Threats	0
■ Low Threats	0
Clean Traffic %	0.0%
Threat Traffic %	100.0%
Session Duration	00:01:08
ML Model Accuracy	Trained (see ml/models/)

Alert Log (100 entries)

Time	Severity	Source	Destination	Attack	Confidence
20:01:04	CRITICAL	192.168.1.13:26763	192.168.1.1:3306	EXPLOIT	51.3%
20:01:04	CRITICAL	192.168.1.14:28983	192.168.1.1:3306	EXPLOIT	51.0%
20:01:03	CRITICAL	192.168.1.21:19995	192.168.1.1:3306	EXPLOIT	51.0%
20:01:03	CRITICAL	192.168.1.4:8906	192.168.1.1:22	EXPLOIT	64.4%
20:01:03	CRITICAL	192.168.1.2:45875	192.168.1.1:443	EXPLOIT	63.9%
20:01:02	CRITICAL	192.168.1.7:6739	192.168.1.1:53	EXPLOIT	63.9%
20:01:02	CRITICAL	192.168.1.28:3166	192.168.1.1:22	EXPLOIT	64.7%
20:01:02	CRITICAL	192.168.1.29:29210	192.168.1.1:80	EXPLOIT	64.2%
20:01:01	CRITICAL	192.168.1.10:27939	192.168.1.1:443	EXPLOIT	63.9%
20:01:01	CRITICAL	192.168.1.26:49961	192.168.1.1:80	EXPLOIT	63.9%
20:01:01	CRITICAL	192.168.1.28:56965	192.168.1.1:53	EXPLOIT	63.9%
20:01:00	CRITICAL	192.168.1.9:1807	192.168.1.1:22	EXPLOIT	58.4%
20:01:00	CRITICAL	192.168.1.15:53819	192.168.1.1:80	EXPLOIT	63.9%
20:01:00	CRITICAL	192.168.1.23:16891	192.168.1.1:443	EXPLOIT	64.2%
20:00:59	CRITICAL	192.168.1.28:62555	192.168.1.1:3306	EXPLOIT	44.8%
20:00:59	CRITICAL	192.168.1.6:37298	192.168.1.1:443	EXPLOIT	58.4%
20:00:59	CRITICAL	192.168.1.5:10019	192.168.1.1:80	EXPLOIT	64.2%
20:00:58	CRITICAL	192.168.1.11:37669	192.168.1.1:25	EXPLOIT	64.7%
20:00:58	CRITICAL	192.168.1.14:51282	192.168.1.1:53	EXPLOIT	63.9%
20:00:57	CRITICAL	192.168.1.26:58386	192.168.1.1:80	EXPLOIT	58.4%
20:00:57	CRITICAL	192.168.1.3:52929	192.168.1.1:22	EXPLOIT	64.4%
20:00:57	CRITICAL	192.168.1.28:16594	192.168.1.1:22	EXPLOIT	64.7%
20:00:56	CRITICAL	192.168.1.24:46067	192.168.1.1:25	EXPLOIT	64.4%
20:00:56	CRITICAL	192.168.1.29:43921	192.168.1.1:25	EXPLOIT	64.7%
20:00:56	CRITICAL	192.168.1.8:6560	192.168.1.1:80	EXPLOIT	63.4%
20:00:56	CRITICAL	192.168.1.3:30953	192.168.1.1:3306	EXPLOIT	51.3%
20:00:55	CRITICAL	192.168.1.2:16935	192.168.1.1:80	EXPLOIT	63.9%
20:00:55	CRITICAL	192.168.1.20:41560	192.168.1.1:80	EXPLOIT	63.4%
20:00:55	CRITICAL	192.168.1.23:31073	192.168.1.1:443	EXPLOIT	63.9%
20:00:54	CRITICAL	192.168.1.7:8700	192.168.1.1:25	EXPLOIT	64.7%
20:00:54	CRITICAL	192.168.1.10:35724	192.168.1.1:80	EXPLOIT	64.2%
20:00:53	CRITICAL	192.168.1.9:55459	192.168.1.1:3306	EXPLOIT	51.0%
20:00:53	CRITICAL	192.168.1.5:36913	192.168.1.1:3306	EXPLOIT	51.3%
20:00:53	CRITICAL	192.168.1.11:24172	192.168.1.1:22	EXPLOIT	64.4%
20:00:52	CRITICAL	192.168.1.20:12725	192.168.1.1:22	EXPLOIT	64.4%
20:00:52	CRITICAL	192.168.1.23:59172	192.168.1.1:80	EXPLOIT	63.9%
20:00:52	CRITICAL	192.168.1.4:18427	192.168.1.1:22	EXPLOIT	64.4%
20:00:51	CRITICAL	192.168.1.15:51743	192.168.1.1:3306	EXPLOIT	51.0%

20:00:51	CRITICAL	192.168.1.5:29657	192.168.1.1:22	EXPLOIT	64.7%
20:00:51	CRITICAL	192.168.1.13:35048	192.168.1.1:25	EXPLOIT	64.7%
20:00:50	CRITICAL	192.168.1.20:46086	192.168.1.1:80	EXPLOIT	58.4%
20:00:50	CRITICAL	192.168.1.11:52928	192.168.1.1:53	EXPLOIT	63.9%
20:00:50	CRITICAL	192.168.1.26:36755	192.168.1.1:3306	EXPLOIT	51.0%
20:00:49	CRITICAL	192.168.1.24:3015	192.168.1.1:443	EXPLOIT	64.2%
20:00:49	CRITICAL	192.168.1.23:18330	192.168.1.1:53	EXPLOIT	64.2%
20:00:49	CRITICAL	192.168.1.24:55005	192.168.1.1:53	EXPLOIT	63.9%
20:00:48	CRITICAL	192.168.1.29:3344	192.168.1.1:53	EXPLOIT	63.9%
20:00:48	CRITICAL	192.168.1.6:48162	192.168.1.1:22	EXPLOIT	64.7%
20:00:48	CRITICAL	192.168.1.6:4492	192.168.1.1:3306	EXPLOIT	51.3%
20:00:47	CRITICAL	192.168.1.16:12908	192.168.1.1:3306	EXPLOIT	51.0%
20:00:47	CRITICAL	192.168.1.8:22215	192.168.1.1:53	EXPLOIT	58.4%
20:00:46	CRITICAL	192.168.1.22:56185	192.168.1.1:25	EXPLOIT	64.4%
20:00:46	CRITICAL	192.168.1.13:58979	192.168.1.1:443	EXPLOIT	63.4%
20:00:46	CRITICAL	192.168.1.12:25370	192.168.1.1:443	EXPLOIT	58.4%
20:00:45	CRITICAL	192.168.1.8:21014	192.168.1.1:25	EXPLOIT	64.4%
20:00:45	CRITICAL	192.168.1.17:60440	192.168.1.1:443	EXPLOIT	63.7%
20:00:44	CRITICAL	192.168.1.5:14980	192.168.1.1:80	EXPLOIT	64.2%
20:00:44	CRITICAL	192.168.1.19:30096	192.168.1.1:22	EXPLOIT	64.7%
20:00:44	CRITICAL	192.168.1.2:2766	192.168.1.1:53	EXPLOIT	64.2%
20:00:43	CRITICAL	192.168.1.8:16974	192.168.1.1:25	EXPLOIT	62.2%
20:00:43	CRITICAL	192.168.1.21:6148	192.168.1.1:443	EXPLOIT	64.2%
20:00:42	CRITICAL	192.168.1.23:6930	192.168.1.1:53	EXPLOIT	63.9%
20:00:42	CRITICAL	192.168.1.13:52256	192.168.1.1:25	EXPLOIT	58.4%
20:00:41	CRITICAL	192.168.1.4:62593	192.168.1.1:22	EXPLOIT	64.4%
20:00:41	CRITICAL	192.168.1.9:56454	192.168.1.1:53	EXPLOIT	63.9%
20:00:40	CRITICAL	192.168.1.24:61966	192.168.1.1:443	EXPLOIT	63.9%
20:00:40	CRITICAL	192.168.1.21:64032	192.168.1.1:25	EXPLOIT	64.7%
20:00:39	CRITICAL	192.168.1.17:5566	192.168.1.1:80	EXPLOIT	64.2%
20:00:39	CRITICAL	192.168.1.12:52071	192.168.1.1:443	EXPLOIT	64.2%
20:00:38	CRITICAL	192.168.1.5:63050	192.168.1.1:25	EXPLOIT	64.4%
20:00:38	CRITICAL	192.168.1.9:26651	192.168.1.1:22	EXPLOIT	64.7%
20:00:38	CRITICAL	192.168.1.24:27209	192.168.1.1:22	EXPLOIT	64.4%
20:00:37	CRITICAL	192.168.1.8:16256	192.168.1.1:22	EXPLOIT	63.9%
20:00:37	CRITICAL	192.168.1.7:36422	192.168.1.1:80	EXPLOIT	64.2%
20:00:37	CRITICAL	192.168.1.29:5730	192.168.1.1:53	EXPLOIT	64.2%
20:00:36	CRITICAL	192.168.1.26:14047	192.168.1.1:3306	EXPLOIT	52.5%
20:00:36	CRITICAL	192.168.1.3:49928	192.168.1.1:443	EXPLOIT	64.2%
20:00:36	CRITICAL	192.168.1.13:55129	192.168.1.1:22	EXPLOIT	39.4%
20:00:35	CRITICAL	192.168.1.15:40874	192.168.1.1:3306	EXPLOIT	33.0%

20:00:35	CRITICAL	192.168.1.26:50458	192.168.1.1:3306	EXPLOIT	34.5%
20:00:34	CRITICAL	192.168.1.20:14368	192.168.1.1:443	EXPLOIT	38.9%
20:00:34	CRITICAL	192.168.1.4:23856	192.168.1.1:443	EXPLOIT	38.9%
20:00:34	CRITICAL	192.168.1.20:41069	192.168.1.1:3306	EXPLOIT	34.5%
20:00:33	CRITICAL	192.168.1.6:8002	192.168.1.1:22	EXPLOIT	39.4%
20:00:33	CRITICAL	192.168.1.26:21032	192.168.1.1:3306	EXPLOIT	34.5%
20:00:33	CRITICAL	192.168.1.28:34475	192.168.1.1:53	EXPLOIT	38.9%
20:00:32	CRITICAL	192.168.1.21:8676	192.168.1.1:53	EXPLOIT	35.4%
20:00:32	CRITICAL	192.168.1.19:15316	192.168.1.1:25	EXPLOIT	39.4%
20:00:32	CRITICAL	192.168.1.11:42885	192.168.1.1:22	EXPLOIT	39.4%
20:00:31	CRITICAL	192.168.1.3:20182	192.168.1.1:3306	EXPLOIT	34.5%
20:00:31	CRITICAL	192.168.1.7:49769	192.168.1.1:80	EXPLOIT	36.9%
20:00:31	CRITICAL	192.168.1.9:16822	192.168.1.1:3306	EXPLOIT	34.5%
20:00:30	CRITICAL	192.168.1.27:57341	192.168.1.1:443	EXPLOIT	38.9%
20:00:30	CRITICAL	192.168.1.7:49334	192.168.1.1:80	EXPLOIT	38.9%
20:00:29	CRITICAL	192.168.1.27:2238	192.168.1.1:80	EXPLOIT	38.9%
20:00:29	CRITICAL	192.168.1.15:57304	192.168.1.1:443	EXPLOIT	38.9%
20:00:28	CRITICAL	192.168.1.7:52219	192.168.1.1:80	EXPLOIT	38.9%
20:00:28	CRITICAL	192.168.1.12:43427	192.168.1.1:53	EXPLOIT	38.9%
20:00:28	CRITICAL	192.168.1.9:38539	192.168.1.1:53	EXPLOIT	38.9%
20:00:27	CRITICAL	192.168.1.22:35547	192.168.1.1:3306	EXPLOIT	34.5%

Top Attacker IPs

Rank	IP Address	Alert Count	Threat Level
1	192.168.1.20	6	HIGH
2	192.168.1.9	6	HIGH
3	192.168.1.7	6	HIGH
4	192.168.1.26	6	HIGH
5	192.168.1.28	5	MEDIUM

MITRE ATT&CK; Framework Mapping

Each detected attack class is mapped to its corresponding MITRE ATT&CK; Tactics and Techniques (Framework v14). This enables security teams to correlate NIDS detections with the broader threat intelligence ecosystem.

Attack Class	Tactic	Technique ID	Technique Name	Severity
DOS	Impact	T1498	Network Denial of Service	CRITICAL
DOS	Impact	T1499	Endpoint Denial of Service	CRITICAL
DDOS	Impact	T1498	Network Denial of Service	CRITICAL
DDOS	Impact	T1499	Endpoint Denial of Service	CRITICAL
PORT_SCAN	Discovery	T1046	Network Service Discovery	MEDIUM
PORT_SCAN	Reconnaissance	T1595	Active Scanning	MEDIUM
PORT_SCAN	Reconnaissance	T1590	Gather Victim Network Information	MEDIUM
VULN_SCAN	Reconnaissance	T1595	Active Scanning	HIGH
VULN_SCAN	Reconnaissance	T1590	Gather Victim Network Information	HIGH
VULN_SCAN	Discovery	T1046	Network Service Discovery	HIGH
BRUTE_FORCE	Credential Access	T1110	Brute Force	HIGH
BRUTE_FORCE	Initial Access	T1021	Remote Services	HIGH
BRUTE_FORCE	Initial Access	T1078	Valid Accounts	HIGH
EXPLOIT	Privilege Escalation	T1068	Exploitation for Privilege Escalation	CRITICAL
EXPLOIT	Privilege Escalation	T1055	Process Injection	CRITICAL
EXPLOIT	Privilege Escalation	T1059	Command and Scripting Interpreter	CRITICAL
EXPLOIT	Privilege Escalation	T1548	Abuse Elevation Control Mechanism	CRITICAL
WEB_ATTACK	Initial Access	T1190	Exploit Public-Facing Application	HIGH
WEB_ATTACK	Privilege Escalation	T1059	Command and Scripting Interpreter	HIGH
WEB_ATTACK	Lateral Movement	T1210	Exploitation of Remote Services	HIGH
INFILTRATION	Command and Control	T1071	Application Layer Protocol	CRITICAL
INFILTRATION	Command and Control	T1105	Ingress Tool Transfer	CRITICAL
INFILTRATION	Initial Access	T1021	Remote Services	CRITICAL
INFILTRATION	Initial Access	T1133	External Remote Services	CRITICAL
EXFILTRATION	Exfiltration	T1041	Exfiltration Over C2 Channel	CRITICAL
EXFILTRATION	Exfiltration	T1048	Exfiltration Over Alternative Protocol	CRITICAL
EXFILTRATION	Exfiltration	T1567	Exfiltration Over Web Service	CRITICAL

Full details at: <https://attack.mitre.org>

Attack Class Summaries

DOS — Impact (TA0040)

Single-source DoS floods targeting network bandwidth or service availability. Detected via packet_rate, error_rate and count anomalies.

ID	Technique	Description (excerpt)
T1498	Network Denial of Service	Adversaries flood network infrastructure to degrade or block availability for target users...
T1499	Endpoint Denial of Service	Adversaries perform DoS attacks to degrade or block availability of targeted resources at ...

DDOS — Impact (TA0040)

Distributed multi-source flooding — significantly higher packet volume than single-source DoS. Amplification and reflection techniques are common.

ID	Technique	Description (excerpt)
T1498	Network Denial of Service	Adversaries flood network infrastructure to degrade or block availability for target users...
T1499	Endpoint Denial of Service	Adversaries perform DoS attacks to degrade or block availability of targeted resources at ...

PORT_SCAN — Discovery (TA0007) + Reconnaissance (TA0043)

Sequential port probing using Nmap, ipsweep, or portsweep to discover open services. Detected via high diff_srv_rate and RST ratio with low bytes.

ID	Technique	Description (excerpt)
T1046	Network Service Discovery	Adversaries scan victim networks to discover services running on remote hosts. Tools like ...
T1595	Active Scanning	Adversaries actively probe victim infrastructure using port scans, vulnerability scans, an...
T1590	Gather Victim Network Information	Adversaries gather information about the target's network infrastructure including IP rang...

VULN_SCAN — Reconnaissance (TA0043)

Targeted vulnerability enumeration against specific services using tools like Satan, Nessus, or OpenVAS. More selective than port scanning.

ID	Technique	Description (excerpt)
T1595	Active Scanning	Adversaries actively probe victim infrastructure using port scans, vulnerability scans, an...
T1590	Gather Victim Network Information	Adversaries gather information about the target's network infrastructure including IP rang...
T1046	Network Service Discovery	Adversaries scan victim networks to discover services running on remote hosts. Tools like ...

BRUTE_FORCE — Credential Access (TA0006)

Repeated authentication attempts against SSH, FTP, or HTTP. Detected via high same_srv_rate, error_rate, and repeated connections to auth ports 21/22/80.

ID	Technique	Description (excerpt)
T1110	Brute Force	Adversaries use brute force techniques to attempt access to accounts when passwords are un...
T1021	Remote Services	Adversaries use valid credentials to log into a service specifically designed to accept re...
T1078	Valid Accounts	Adversaries obtain and abuse credentials of existing accounts as a means of gaining initia...

EXPLOIT — Privilege Escalation (TA0004)

Buffer overflow, shellcode injection, or kernel module exploitation to gain root/SYSTEM access. Detected via very high src_bytes, long duration, and urgent flags.

ID	Technique	Description (excerpt)
T1068	Exploitation for Privilege Escalation	Adversaries exploit software vulnerabilities in an attempt to elevate privileges. This is ...
T1055	Process Injection	Adversaries inject code into processes to evade process-based defences and possibly elevat...
T1059	Command and Scripting Interpreter	Adversaries abuse command and script interpreters to execute commands, scripts, or binarie...
T1548	Abuse Elevation Control Mechanism	Adversaries bypass mechanisms designed to control elevate privileges to perform privileged...

WEB_ATTACK — Initial Access (TA0001) + Execution (TA0002)

SQL injection, XSS, or HTTP exploitation targeting web applications on port 80/443. Detected via crafted payloads in HTTP requests with unusual byte patterns.

ID	Technique	Description (excerpt)
T1190	Exploit Public-Facing Application	Adversaries exploit weaknesses in internet-facing software including web applications, dat...
T1059	Command and Scripting Interpreter	Adversaries abuse command and script interpreters to execute commands, scripts, or binarie...
T1210	Exploitation of Remote Services	Adversaries exploit remote services to gain access to internal systems. This includes expl...

INFILTRATION — Command and Control (TA0011) + Lateral Movement (TA0008)

Persistent backdoor or C2 channel maintaining covert access. Detected via long session duration on non-standard ports with low but steady traffic patterns.

ID	Technique	Description (excerpt)
T1071	Application Layer Protocol	Adversaries communicate using application layer protocols to avoid detection. HTTP, HTTPS,...
T1105	Ingress Tool Transfer	Adversaries transfer tools or malware from external systems into a compromised network env...
T1021	Remote Services	Adversaries use valid credentials to log into a service specifically designed to accept re...

T1133	External Remote Services	Adversaries leverage external-facing remote services such as VPN, Citrix, RDP, and SSH to ...
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EXFILTRATION — Exfiltration (TA0010)

Large unauthorised outbound data transfer via FTP, HTTP, or covert channels. Detected via abnormally high `dst_bytes` and `byte_rate` on standard ports.

ID	Technique	Description (excerpt)
T1041	Exfiltration Over C2 Channel	Adversaries steal data over the existing C2 channel. This avoids creating new outbound con...
T1048	Exfiltration Over Alternative Protocol	Adversaries use protocols other than the C2 channel to exfiltrate data — DNS tunnelling, I...
T1567	Exfiltration Over Web Service	Adversaries use legitimate web services (Google Drive, Dropbox, GitHub, Pastebin) to exfil...

AI Explanation Samples (Top 3 Alerts)

Alert: EXPLOIT from 192.168.1.13:26763 — Confidence: 51.3%

Feature	Value	Contribution	Direction
wrong_fragment	0.0	35.8%	↓ decreases risk
srv_count	1.0	23.4%	↑ increases risk
urgent	0.0	12.7%	↓ decreases risk
count	1.0	10.6%	↑ increases risk
src_bytes	492.0	9.4%	↑ increases risk

Alert: EXPLOIT from 192.168.1.14:28983 — Confidence: 51.0%

Feature	Value	Contribution	Direction
wrong_fragment	0.0	37.9%	↓ decreases risk
srv_count	1.0	23.3%	↑ increases risk
count	1.0	10.6%	↑ increases risk
urgent	0.0	10.4%	↓ decreases risk
src_bytes	229.0	9.3%	↑ increases risk

Alert: EXPLOIT from 192.168.1.21:19995 — Confidence: 51.0%

Feature	Value	Contribution	Direction
wrong_fragment	0.0	38.0%	↓ decreases risk
srv_count	1.0	23.3%	↑ increases risk
count	1.0	10.7%	↑ increases risk
urgent	0.0	10.5%	↓ decreases risk
src_bytes	126.0	9.4%	↑ increases risk

Recommendations

1. Block attacker IPs

Use firewall rules to block the top source IPs identified in the alert log.

2. Investigate DoS flows

High packet-rate flows from single sources should be traced to origin. Consider rate limiting at the network edge.

3. Review probe targets

Port scan targets indicate which services are being enumerated. Disable unused services and apply strict firewall ingress rules.

4. Patch R2L vulnerabilities

Remote-to-Local attack attempts suggest exposed services (SSH, FTP, SMTP). Enforce key-based auth and disable password login.

5. Enable continuous monitoring

Deploy NIDS in persistent mode with SIEM integration and automated alerting for 24/7 coverage.

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